

An algorithm for SOI encounter detection on Keplerian orbits

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1. Problem Statement

Given two Keplerian orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI} . It is possible that such E_1, E_2 don't exist, the algorithm needs to handle this case as well. The spacecraft orbit may be elliptic ($e_1 < 1$) or hyperbolic ($e_1 > 1$); the planet orbit is always elliptic.

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where A and B are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

We divide this hard problem into two easier ones:

1. Finding tight **ranges** of E_1, E_2 **where** encounters are **geometrically** possible, ignoring real positions of the bodies
2. Finding precise **values** of E_1 and time t **when** the spacecraft **actually** enters sphere-of-influence of the planet

2. Position in Orbital Plane

For an elliptic orbit, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{bmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{bmatrix} \quad (2)$$

For a hyperbolic spacecraft orbit:

$$\mathbf{r}_1(H_1) = \begin{bmatrix} a_1(\cosh H_1 - e_1) \\ b_1 \sinh H_1 \end{bmatrix} \quad (3)$$

3. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (4)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (5)$$

3.1. Orthonormality of A and B

Since A and B are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (6)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (7)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (8)$$

3.2. Coupling matrix C

Define the 2×2 matrix:

$$C := 2A^\top B \quad (9)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 5 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (10)$$

Note: C is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

3.3. Scaled coupling matrix M

The position vector Equation 2 can be factored as $\mathbf{r}_i = \text{diag}(a_i, b_i)\mathbf{p}_i$, where $\text{diag}(a_i, b_i) := \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}$ is a diagonal matrix and $\mathbf{p}_i$ is a dimensionless position-direction vector:

$$\mathbf{p}_i = \begin{bmatrix} \cos E_i - e_i \\ \sin E_i \end{bmatrix} \quad (11)$$

For a hyperbolic spacecraft orbit:

$$\mathbf{p}_1 = \begin{bmatrix} \cosh H_1 - e_1 \\ \sinh H_1 \end{bmatrix} \quad (12)$$

Substituting $\mathbf{r}_i = \text{diag}(a_i, b_i)\mathbf{p}_i$ into the cross-term and using $(\mathbf{X}\mathbf{p})^\top = \mathbf{p}^\top \mathbf{X}^\top$:

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = (\text{diag}(a_1, b_1)\mathbf{p}_1)^\top \mathbf{C} \text{diag}(a_2, b_2)\mathbf{p}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1)^\top \mathbf{C} \text{diag}(a_2, b_2)\mathbf{p}_2 \quad (13)$$

Since diagonal matrices are symmetric ($\text{diag}(x, y)^\top = \text{diag}(x, y)$):

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2)\mathbf{p}_2 \quad (14)$$

We have a chain of five matrices. Applying associativity $(\mathbf{A}(\mathbf{B}\mathbf{C})) = (\mathbf{A}\mathbf{B})\mathbf{C}$ twice – first to $(\mathbf{p}_1^\top \cdot \text{diag}(a_1, b_1)) \cdot \mathbf{C} = \mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})$, then to $(\mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})) \cdot \text{diag}(a_2, b_2) = \mathbf{p}_1^\top \cdot ((\text{diag}(a_1, b_1) \cdot \mathbf{C}) \cdot \text{diag}(a_2, b_2))$ – lets us group the three constant middle factors into a single **scaled coupling matrix** $\mathbf{M}$.

$$\mathbf{M} = \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) \quad (15)$$

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 \quad (16)$$

Let's introduce matrix $\mathbf{D}$ and express $\mathbf{M}$ in terms of it ($\circ$ denotes element-wise product):

$$\mathbf{D} := \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}^\top = \begin{pmatrix} a_1 a_2 & a_1 b_2 \\ b_1 a_2 & b_1 b_2 \end{pmatrix}, \quad \mathbf{M} := \mathbf{D} \circ \mathbf{C} \quad (17)$$

4. Coarse encounter intervals

We restrict the search to eccentric anomaly values where the spacecraft's heliocentric distance lies between planetary perihelion and aphelion $\pm r_{\text{SOI}}$.

4.1. Radial overlap condition

An encounter requires the spacecraft distance to lie within:

$$a_2(1 - e_2) - r_{\text{SOI}} \leq r_1 \leq a_2(1 + e_2) + r_{\text{SOI}} \quad (18)$$

Elliptic spacecraft. Using $r_1 = a_1(1 - e_1 \cos E_1)$ and solving for $\cos E_1$:

$$\frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \leq \cos E_1 \leq \frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \quad (19)$$

Clamping both sides to $[-1, 1]$ and applying $\arccos$ (which reverses the inequality) gives two symmetric intervals $[E_{\text{lo}}, E_{\text{hi}}]$ and $[-E_{\text{lo}}, -E_{\text{hi}}]$ in E_1 . If no valid interval exists (i.e. the clamped range is empty), no encounter is possible at any E_1 .

Hyperbolic spacecraft. Using $r_1 = a_1(1 - e_1 \cosh H_1)$ and solving for $\cosh H_1$ (noting $a_1 e_1 < 0$, so dividing reverses the inequality):

$$\frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \leq \cosh H_1 \leq \frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \quad (20)$$

Clamping the lower bound to $\max(\dots, 1)$ (since $\cosh H_1 \geq 1$) and applying arcosh gives a single symmetric interval $[-H_{\text{hi}}, -H_{\text{lo}}] \cup [H_{\text{lo}}, H_{\text{hi}}]$. If the clamped range is empty, no encounter is possible.

4.2. Initial guess via parabolic interpolation

For each of the two symmetric E_1 intervals, we evaluate f at three points: the endpoints and midpoint. Here E_2 is estimated by projecting the spacecraft's 3D position onto the planet's orbital plane and inverting the ellipse parametrisation:

$$E_2 = \text{atan2}(q_y/b_2, q_x/a_2 + e_2) \quad (21)$$

where $\mathbf{q} = \mathbf{B}^\top \mathbf{A} \mathbf{r}_1$ is the projection. This matches the eccentric anomaly that $\mathbf{q}$ would have if it lay on the planet's ellipse. The projection is exact when the orbital planes coincide; for inclined orbits, the out-of-plane component is lost. Numerical evaluation for a Pluto-like target ($i = 17^\circ$, $e_2 = 0.25$) shows the maximum E_2 error is under 2° for moderate spacecraft eccentricities ($e_1 < 0.5$), but a direct Earth–Pluto near-Hohmann transfer ($e_1 \approx 0.95$) reaches $\sim 9^\circ$ at favourable orientations and up to $\sim 63^\circ$ at unfavourable ascending-node orientations. Despite these large worst-case errors, the estimate remains adequate as an initial guess for Newton's method, which converges from any starting point in the correct half of the encounter region.

Given three samples (x_0, f_0) , (x_1, f_1) , (x_2, f_2) , the minimiser of the interpolating quadratic is:

$$E_1^* = x_1 - \frac{1}{2} \frac{(x_1 - x_0)^2(f_1 - f_2) - (x_1 - x_2)^2(f_1 - f_0)}{(x_1 - x_0)(f_1 - f_2) - (x_1 - x_2)(f_1 - f_0)} \quad (22)$$

Newton's method is then run from the best initial guess across both branches. If the first branch fails to converge to $f \leq r_{\text{SOI}}^2$, the second branch is tried.

5. Finding minimum distance using Newton's Method

We seek to minimise $f(E_1, E_2) := r_1^2 + r_2^2 - \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 - r_{\text{SOI}}^2$ using Newton's method. Each iteration solves the 2×2 linear system $\mathbf{H} \delta = -\nabla f$ for the step $\delta = (\delta E_1, \delta E_2)^\top$, then updates $E_i \leftarrow E_i + \delta E_i$.

5.1. Deriving the gradient and Hessian

We differentiate each term with respect to E_i .

5.1.1. Self-terms r_i^2

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \quad (23)$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2a_i^2 e_i (\cos E_i - e_i + 2e_i \sin^2 E_i) \quad (24)$$

and $\partial r_i^2 / \partial E_j = 0$ for $i \neq j$.

For a hyperbolic spacecraft orbit, $r_1 = a_1(1 - e_1 \cosh H_1)$ (positive since $a_1 < 0$ and $1 - e_1 \cosh H_1 < 0$):

$$\frac{\partial r_1^2}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) \quad (25)$$

$$\frac{\partial^2 r_1^2}{\partial H_1^2} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) \quad (26)$$

5.1.2. Cross-term $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$

Since $\mathbf{M}$ is constant, differentiation acts only on $\mathbf{p}_1$ and $\mathbf{p}_2$. Define the derivative vectors:

$$\hat{\mathbf{w}}_i := \frac{d\mathbf{p}_i}{dE_i} = \begin{bmatrix} -\sin E_i \\ \cos E_i \end{bmatrix}, \quad \hat{\mathbf{u}}_i := \begin{bmatrix} \cos E_i \\ \sin E_i \end{bmatrix} \quad (27)$$

For a hyperbolic spacecraft orbit:

$$\hat{\mathbf{w}}_1 := \frac{d\mathbf{p}_1}{dH_1} = \begin{bmatrix} \sinh H_1 \\ \cosh H_1 \end{bmatrix}, \quad \hat{\mathbf{u}}_1 := \begin{bmatrix} \cosh H_1 \\ \sinh H_1 \end{bmatrix} \quad (28)$$

To differentiate $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$ with respect to E_1 , note that only $\mathbf{p}_1$ depends on E_1 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = \mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2)$, which is a dot product of $\mathbf{p}_1$ with the constant vector $\mathbf{M} \mathbf{p}_2$. The derivative of a dot product with one constant factor is:

$$\frac{\partial(\mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2))}{\partial E_1} = \hat{\mathbf{w}}_1^\top (\mathbf{M} \mathbf{p}_2) = \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (29)$$

Similarly, for E_2 only $\mathbf{p}_2$ depends on E_2 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = (\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2$, a dot product of the constant row vector $\mathbf{p}_1^\top \mathbf{M}$ with $\mathbf{p}_2$:

$$\frac{\partial((\mathbf{p}_1^\top \mathbf{M})\mathbf{p}_2)}{\partial E_2} = (\mathbf{p}_1^\top \mathbf{M})\hat{\mathbf{w}}_2 = \mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{w}}_2 \quad (30)$$

The same reasoning applies to higher derivatives, replacing $\mathbf{p}_i$ by successive derivatives $\hat{\mathbf{w}}_i$, $-\hat{\mathbf{u}}_i$ (elliptic) or $+\hat{\mathbf{u}}_i$ (hyperbolic, since $d^2\mathbf{p}_1/dH_1^2 = +\hat{\mathbf{u}}_1$):

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M}\mathbf{p}_2)}{\partial E_1^2} = -\hat{\mathbf{u}}_1^\top \mathbf{M}\mathbf{p}_2, \quad \frac{\partial^2(\mathbf{p}_1^\top \mathbf{M}\mathbf{p}_2)}{\partial E_2^2} = -\mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{u}}_2 \quad (31)$$

For a hyperbolic spacecraft orbit, the sign of the E_1 term flips:

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M}\mathbf{p}_2)}{\partial H_1^2} = +\hat{\mathbf{u}}_1^\top \mathbf{M}\mathbf{p}_2 \quad (32)$$

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M}\mathbf{p}_2)}{\partial E_1 \partial E_2} = \hat{\mathbf{w}}_1^\top \mathbf{M}\hat{\mathbf{w}}_2 \quad (33)$$

5.1.3. Gradient

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M}\mathbf{p}_2 \quad (34)$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{w}}_2 \quad (35)$$

For a hyperbolic spacecraft orbit, Equation 34 becomes (using Equation 25 and the hyperbolic $\hat{\mathbf{w}}_1$, $\mathbf{p}_1$ from Equation 28):

$$\frac{\partial f}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) - \hat{\mathbf{w}}_1^\top \mathbf{M}\mathbf{p}_2 \quad (36)$$

Equation 35 is unchanged (the planet is always elliptic), but $\mathbf{p}_1$ and $\mathbf{M}$ use the hyperbolic definitions.

5.1.4. Hessian

The Hessian $\mathbf{H}$ is symmetric ($H_{12} = H_{21}$):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \hat{\mathbf{u}}_1^\top \mathbf{M}\mathbf{p}_2 \quad (37)$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{u}}_2 \quad (38)$$

$$H_{12} = -\hat{\mathbf{w}}_1^\top \mathbf{M}\hat{\mathbf{w}}_2 \quad (39)$$

For a hyperbolic spacecraft orbit, H_{11} changes (using Equation 26 and Equation 32):

$$H_{11} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) - \hat{\mathbf{u}}_1^\top \mathbf{M}\mathbf{p}_2 \quad (40)$$

H_{22} and H_{12} retain the same form, with the hyperbolic $\hat{\mathbf{w}}_1$, $\hat{\mathbf{u}}_1$, $\mathbf{p}_1$.

5.2. Solving the Newton step

The 2×2 system is solved directly via Cramer's rule:

$$\Delta = H_{11}H_{22} - H_{12}^2 \quad (41)$$

$$\delta E_1 = \frac{\frac{\partial f}{\partial E_1} H_{22} - \frac{\partial f}{\partial E_2} H_{12}}{-\Delta}, \quad \delta E_2 = \frac{\frac{\partial f}{\partial E_2} H_{11} - \frac{\partial f}{\partial E_1} H_{12}}{-\Delta} \quad (42)$$

All coupling terms are bilinear forms $\mathbf{x}^\top \mathbf{M}\mathbf{y}$ with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ($\hat{\mathbf{w}}_1^\top \mathbf{M}\mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{w}}_2$, $\hat{\mathbf{u}}_1^\top \mathbf{M}\mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M}\hat{\mathbf{u}}_2$, $\hat{\mathbf{w}}_1^\top \mathbf{M}\hat{\mathbf{w}}_2$) can share the intermediate products $\mathbf{M}\mathbf{p}_2$, $\mathbf{M}\hat{\mathbf{w}}_2$, $\mathbf{M}\hat{\mathbf{u}}_2$ (each a 2D matrix-vector multiply).

6. Encounter Intervals in E_1 and E_2

In previous section we've obtained the minima of the distance function $f(E_1, E_2)$. If the minimum is less than r_{SOI} , then spacecraft enters SOI of the planet. However, our end goal is to find **where** and **when** does the spacecraft intersect SOI **boundary**. In other words, finding roots of $f(E_1, E_2) = 0$.

The constraint $f(E_1, E_2) = 0$ defines an implicit curve (or set of curves) in the (E_1, E_2) plane. The **encounter interval** $[E_{1, \min}, E_{1, \max}]$ is the projection of this curve onto the E_1 axis, and $[E_{2, \min}, E_{2, \max}]$ is the projection onto the E_2 axis. For every E_1 in the first interval there exists at least one E_2 in the second interval such that the distance equals r_{SOI} .

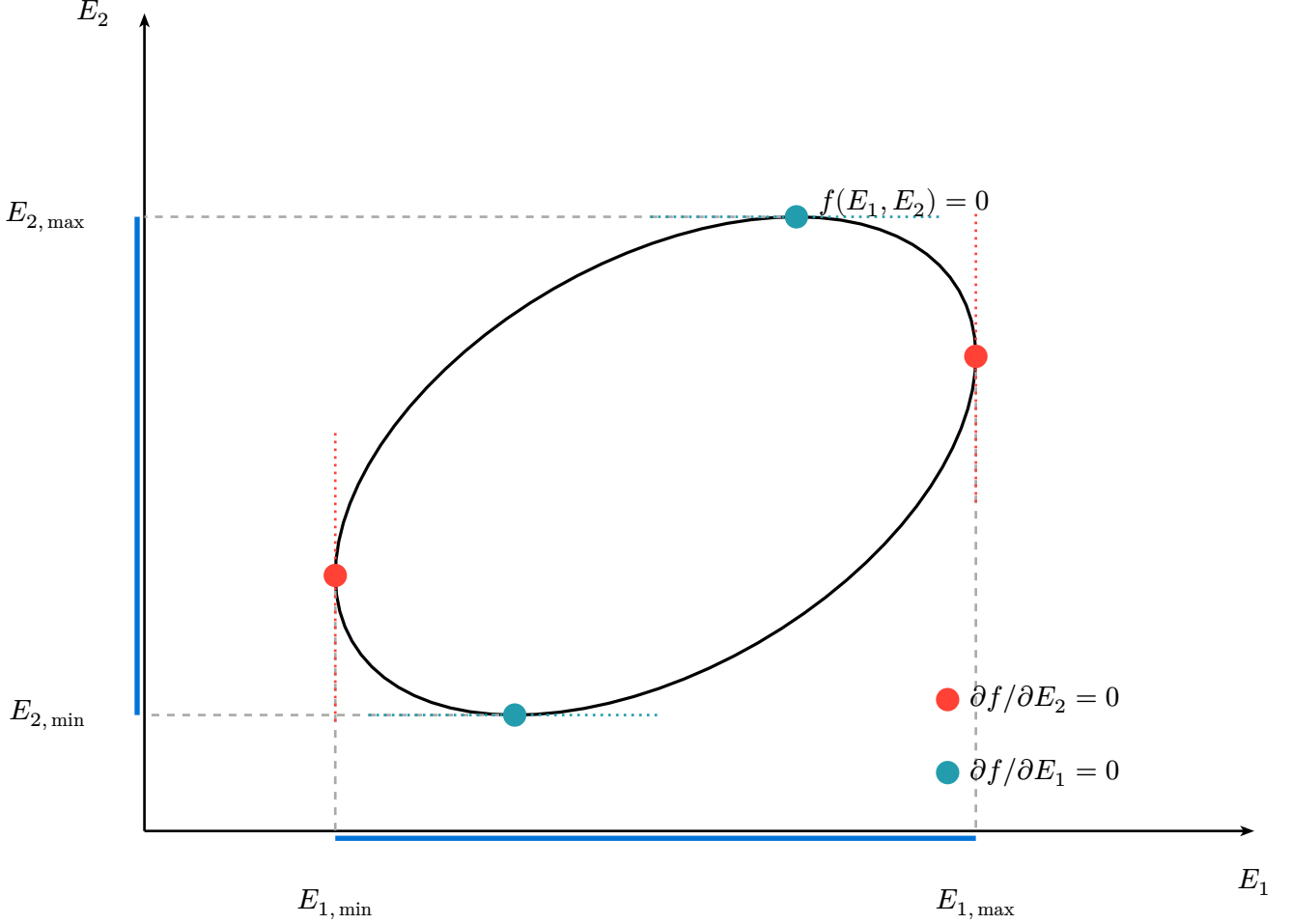


Figure 1: The constraint curve $f = 0$ in the (E_1, E_2) plane. Red dots mark vertical tangents ($\partial f / \partial E_2 = 0$), whose E_1 coordinates give $E_{1, \min}$ and $E_{1, \max}$. Teal dots mark horizontal tangents ($\partial f / \partial E_1 = 0$), whose E_2 coordinates give $E_{2, \min}$ and $E_{2, \max}$. Blue bars on the axes show the projected encounter intervals.

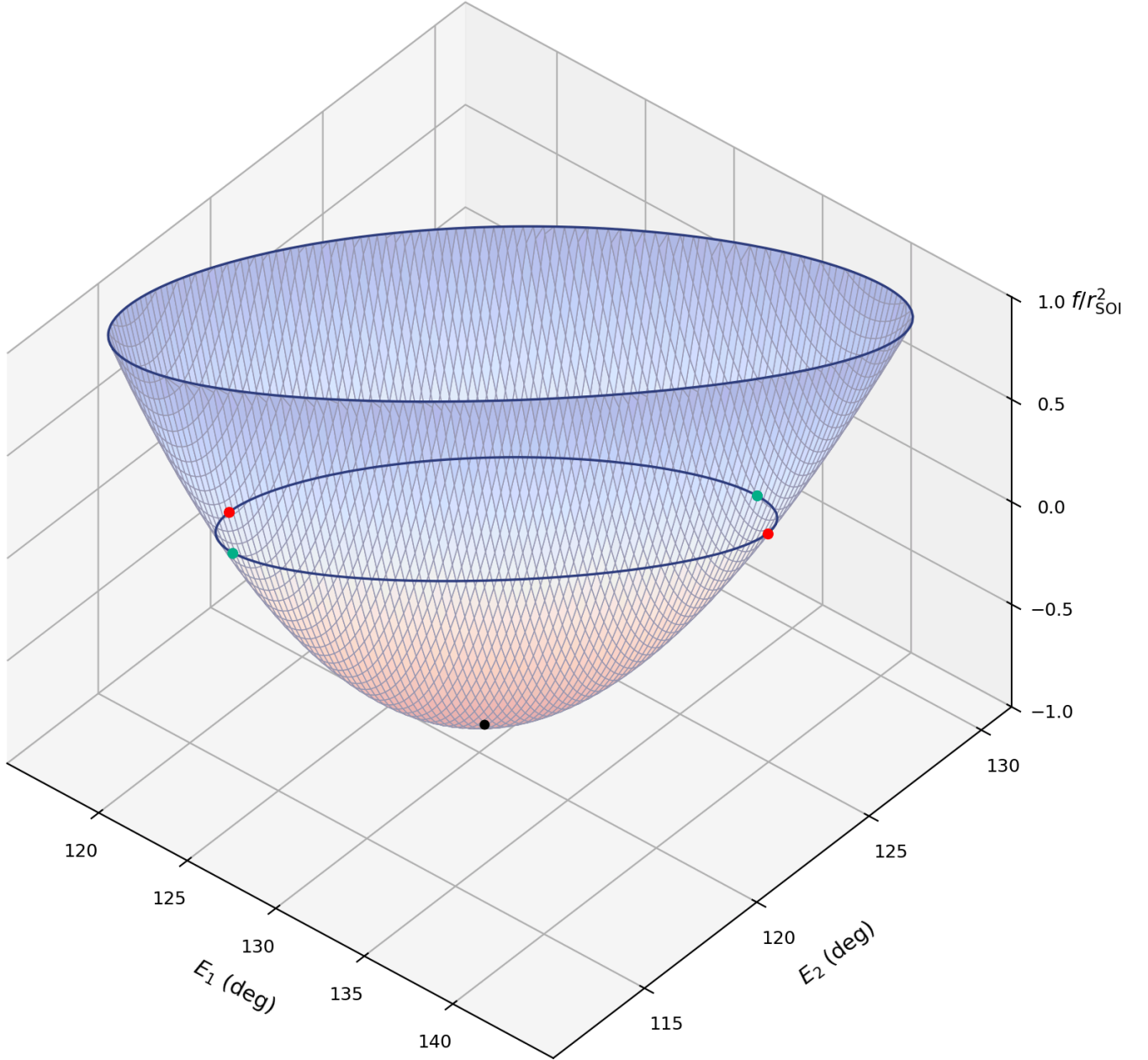


Figure 2: The surface $z = f(E_1, E_2)/r_{\text{SOI}}^2$ over the encounter region (Voyager 2 – Jupiter). The black curve is the $f = 0$ contour where the surface crosses the $z = 0$ plane. The black dot at $(E_1, E_2) \approx (128.9^\circ, 121.1^\circ)$ marks the surface minimum (closest approach without accounting for the planet’s gravity). Red dots mark $\partial f/\partial E_2 = 0$ at $E_1 \approx 120.2^\circ$ and 139.0° (E_1 extrema); green dots mark $\partial f/\partial E_1 = 0$ at $E_2 \approx 115.3^\circ$ and 126.9° (E_2 extrema).

6.1. Extrema via Lagrange multipliers

Finding the extreme values of E_2 subject to $f(E_1, E_2) = 0$ is a constrained optimisation problem. The Lagrangian is:

$$\mathcal{L} = E_2 - \lambda f(E_1, E_2) \quad (43)$$

Setting partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial E_1} = -\lambda \frac{\partial f}{\partial E_1} = 0, \quad \frac{\partial \mathcal{L}}{\partial E_2} = 1 - \lambda \frac{\partial f}{\partial E_2} = 0 \quad (44)$$

The second equation gives $\lambda = 1/(\partial f/\partial E_2)$, which is necessarily nonzero. Substituting into the first:

$$-\frac{1}{\partial f/\partial E_2} \cdot \frac{\partial f}{\partial E_1} = 0 \quad (45)$$

Since the prefactor is nonzero, this requires:

$$\frac{\partial f}{\partial E_1} = 0 \quad (46)$$

By the same argument with $\mathcal{L} = E_1 - \lambda f$, the extrema of E_1 require:

$$\frac{\partial f}{\partial E_2} = 0 \quad (47)$$

Geometrically, consider the surface $z = f(E_1, E_2)$ over the (E_1, E_2) plane (Figure 2). The constraint curve $f = 0$ is the intersection of this surface with the $z = 0$ plane. The E_2 extrema of this intersection occur where the curve runs parallel to the E_1 axis — at these points, the surface's gradient in the E_1 direction is zero along the constraint, giving $\partial f / \partial E_1 = 0$.

6.2. System for E_1 bounds

To find $E_{1, \min}$ and $E_{1, \max}$, solve the 2×2 system:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_2} = 0 \end{cases} \quad (48)$$

Using Equation 35, the second equation is:

$$2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 = 0 \quad (49)$$

Newton's method for this system uses the Jacobian:

$$\mathbf{J}_1 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1 \partial E_2} & \frac{\partial^2 f}{\partial E_2^2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{12} & H_{22} \end{pmatrix} \quad (50)$$

where $g_1 := \partial f / \partial E_1$ (Equation 34), $g_2 := \partial f / \partial E_2$ (Equation 35), and H_{12}, H_{22} are the Hessian entries (Equation 39, Equation 38). The Newton step $\delta = (\delta E_1, \delta E_2)^\top$ solves $\mathbf{J}_1 \delta = -(\mathbf{f}, g_2)^\top$ via Cramer's rule:

$$\Delta_1 = g_1 H_{22} - g_2 H_{12} \quad (51)$$

$$\delta E_1 = \frac{f H_{22} - g_2^2}{\Delta_1}, \quad \delta E_2 = \frac{f H_{12} - g_1 g_2}{\Delta_1} \quad (52)$$

Note that at convergence ($g_2 = 0$), the determinant simplifies to $\Delta_1 = g_1 H_{22}$.

6.3. System for E_2 bounds

To find $E_{2, \min}$ and $E_{2, \max}$, solve:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_1} = 0 \end{cases} \quad (53)$$

Using Equation 34, the second equation is:

$$2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 = 0 \quad (54)$$

The Jacobian is:

$$\mathbf{J}_2 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1^2} & \frac{\partial^2 f}{\partial E_1 \partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{11} & H_{12} \end{pmatrix} \quad (55)$$

The Newton step solves $\mathbf{J}_2 \delta = -(\mathbf{f}, g_1)^\top$:

$$\Delta_2 = g_1 H_{12} - g_2 H_{11} \quad (56)$$

$$\delta E_1 = \frac{f H_{12} - g_1 g_2}{\Delta_2}, \quad \delta E_2 = \frac{g_1^2 - f H_{11}}{\Delta_2} \quad (57)$$

At convergence ($g_1 = 0$), the determinant simplifies to $\Delta_2 = -g_2 H_{11}$.

7. Time Constraint via Kepler's Equation

The preceding sections treated E_1 and E_2 as independent variables. In reality, both bodies obey Kepler's equation, which couples each eccentric anomaly to a common time t .

7.1. Kepler's equation

The mean motion $n_i = \sqrt{\mu/|a_i|^3}$ applies to both elliptic and hyperbolic orbits, since the derivation from the vis-viva equation depends only on $|a_i|$.

For an elliptic orbit i with mean anomaly at epoch $M_{i,0}$:

$$M_i(t) = M_{i,0} + n_i t = E_i - e_i \sin E_i \quad (58)$$

For a hyperbolic spacecraft orbit:

$$M_1(t) = M_{1,0} + n_1 t = e_1 \sinh H_1 - H_1 \quad (59)$$

At a given time t , both anomalies are determined:

$$E_1 - e_1 \sin E_1 = M_{1,0} + n_1 t, \quad E_2 - e_2 \sin E_2 = M_{2,0} + n_2 t \quad (60)$$

(replacing the first equation with Equation 59 for a hyperbolic spacecraft).

7.2. Eliminating time

The position on an ellipse is 2π -periodic in E_i , so each body returns to the same point after each full orbit. The set of times at which elliptic orbit i passes through eccentric anomaly $E_i \in [-\pi, \pi]$ is:

$$t = \frac{E_i - e_i \sin E_i - M_{i,0} + 2\pi k_i}{n_i}, \quad k_i \in \mathbb{Z} \quad (61)$$

A hyperbolic orbit is not periodic — the spacecraft passes through each H_1 exactly once:

$$t = \frac{e_1 \sinh H_1 - H_1 - M_{1,0}}{n_1} \quad (62)$$

For an encounter, both bodies must be at their respective positions **simultaneously**. For two elliptic orbits, equating the time expressions:

$$\frac{E_1 - e_1 \sin E_1 - M_{1,0} + 2\pi k_1}{n_1} = \frac{E_2 - e_2 \sin E_2 - M_{2,0} + 2\pi k_2}{n_2} \quad (63)$$

Rearranging gives a family of **time-coupling constraints**, one for each integer pair (k_1, k_2) :

$$h_{k_1, k_2}(E_1, E_2) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi(n_2 k_1 - n_1 k_2) = 0 \quad (64)$$

Each choice of (k_1, k_2) corresponds to a different encounter opportunity (i.e. orbit 1 on its k_1 -th revolution meeting orbit 2 on its k_2 -th revolution). Since only the combination $n_2 k_1 - n_1 k_2$ appears, the distinct constraints are parametrised by a single offset:

$$h(E_1, E_2; \alpha) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi\alpha = 0 \quad (65)$$

where $\alpha = n_2 k_1 - n_1 k_2$ ranges over a discrete set. In the (E_1, E_2) plane (with $E_i \in [-\pi, \pi]$), each value of α gives a monotone curve from bottom-left to top-right, and consecutive curves are spaced apart by one spacecraft orbital period $T_1 = 2\pi/n_1$. The full encounter problem is: find (E_1, E_2) satisfying both the distance constraint $f = 0$ (Equation 10) and $h = 0$ (Equation 65) for some admissible α .

Hyperbolic spacecraft. Since $k_1 = 0$ (Equation 62), the time constraint simplifies to a family parametrised by k_2 alone. Replacing the spacecraft's mean anomaly expression:

$$h(H_1, E_2; k_2) := n_2(e_1 \sinh H_1 - H_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) - 2\pi n_1 k_2 = 0 \quad (66)$$

7.3. Finding the next encounter after t_0

After each gravity assist, the spacecraft's orbit changes, so it only makes sense to find the **next** encounter from a given epoch t_0 rather than enumerating all future encounters on a fixed orbit.

The distance constraint can produce up to two separate encounter zones (corresponding to the two symmetric coarse intervals from Equation 19). Each zone $s \in \{1, 2\}$ has its own encounter intervals $[E_{1,\min}^{(s)}, E_{1,\max}^{(s)}]$ and $[E_{2,\min}^{(s)}, E_{2,\max}^{(s)}]$ from Figure 1. Since the mean anomaly $M_i = E_i - e_i \sin E_i$ is monotone in E_i , each zone corresponds to mean-anomaly intervals:

$$M_{i,\min}^{(s)} = E_{i,\min}^{(s)} - e_i \sin E_{i,\min}^{(s)}, \quad M_{i,\max}^{(s)} = E_{i,\max}^{(s)} - e_i \sin E_{i,\max}^{(s)} \quad (67)$$

For a hyperbolic spacecraft orbit: $M_{1,\min}^{(s)} = e_1 \sinh H_{1,\min}^{(s)} - H_{1,\min}^{(s)}$ (and likewise for $M_{1,\max}^{(s)}$).

On its k_i -th elliptic orbit, body i passes through encounter zone s during the time interval (Equation 61):

$$t \in \left[\frac{M_{i,\min}^{(s)} - M_{i,0} + 2\pi k_i}{n_i}, \frac{M_{i,\max}^{(s)} - M_{i,0} + 2\pi k_i}{n_i} \right] \quad (68)$$

A hyperbolic spacecraft passes through each encounter zone at most once. Setting $k_1 = 0$ in the above (with the hyperbolic mean anomaly from Equation 62):

$$t \in \left[\frac{M_{1,\min}^{(s)} - M_{1,0}}{n_1}, \frac{M_{1,\max}^{(s)} - M_{1,0}}{n_1} \right] \quad (69)$$

An encounter requires both bodies to be in their respective intervals of the **same** zone **simultaneously**.

7.3.1. Choosing the iteration order

Label the body with the tighter encounter zone (smaller $M_{i,\max}^{(s)} - M_{i,\min}^{(s)}$) as i and the other as j . Iterating over passes of the tighter-zone body produces fewer candidates, since each pass occupies a shorter time interval and is less likely to overlap with the other body. The formulas below use i for the outer loop and j for the overlap check. For a hyperbolic spacecraft, always start with the spacecraft ($i = 1$): it has a single fixed time interval per zone (Equation 69), so just find which planet passes (k_2) overlap it.

7.3.2. First pass of body i after t_0

For each zone s , body i enters the encounter zone for the first time after t_0 on orbit number:

$$k_i^{(s)} = \left\lceil \frac{n_i t_0 + M_{i,0} - M_{i,\max}^{(s)}}{2\pi} \right\rceil \quad (70)$$

This is the smallest integer k_i for which the encounter interval of body i (Equation 68) has not ended before t_0 . Start with $k_i^* = \min_s k_i^{(s)}$, the earliest pass that enters **either** zone.

7.3.3. Checking for overlap with body j

On each pass k_i^* , check both zones $s \in \{1, 2\}$. The encounter window of body i in zone s , clamped to start no earlier than t_0 , is:

$$t_{i,\text{lo}}^{(s)} = \max \left(\frac{M_{i,\min}^{(s)} - M_{i,0} + 2\pi k_i^*}{n_i}, t_0 \right), \quad t_{i,\text{hi}}^{(s)} = \frac{M_{i,\max}^{(s)} - M_{i,0} + 2\pi k_i^*}{n_i} \quad (71)$$

If $t_{i,\text{lo}}^{(s)} > t_{i,\text{hi}}^{(s)}$, body i is not in zone s on this pass. Otherwise, the earliest orbit of body j that could overlap is:

$$k_j = \left\lceil \frac{n_j t_{i,\text{lo}}^{(s)} + M_{j,0} - M_{j,\max}^{(s)}}{2\pi} \right\rceil \quad (72)$$

Body j enters zone s on orbit k_j at time $t_{j,\text{lo}} = \frac{M_{j,\min}^{(s)} - M_{j,0} + 2\pi k_j}{n_j}$. If $t_{j,\text{lo}} \leq t_{i,\text{hi}}^{(s)}$, the two windows overlap — this zone produces a candidate encounter. If both zones produce a match on the same pass, pick the one with the smaller $t_{j,\text{lo}}$ (the

earlier encounter). The resulting pair (k_1, k_2) yields $\alpha = n_2 k_1 - n_1 k_2$, which is the value to use in Newton's method on the combined system (Equation 75).

If neither zone produces a match on pass k_i^* , increment $k_i^* \leftarrow k_i^* + 1$ and repeat. For a hyperbolic spacecraft ($i = 1$), there is no incrementing — if no planet pass overlaps the spacecraft's single encounter window, no encounter exists on this orbit.

7.4. Gradient of h

$$\frac{\partial h}{\partial E_1} = n_2(1 - e_1 \cos E_1), \quad \frac{\partial h}{\partial E_2} = -n_1(1 - e_2 \cos E_2) \quad (73)$$

For a hyperbolic spacecraft orbit:

$$\frac{\partial h}{\partial H_1} = n_2(e_1 \cosh H_1 - 1) \quad (74)$$

This is always positive (since $e_1 > 1$ and $\cosh H_1 \geq 1$), matching the sign of the elliptic case.

7.5. Newton's method on the combined system

For a given α , we solve the 2×2 system $(f, h) = \mathbf{0}$ using Newton's method. The Jacobian is:

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial h}{\partial E_1} & \frac{\partial h}{\partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ n_2(1 - e_1 \cos E_1) & -n_1(1 - e_2 \cos E_2) \end{pmatrix} \quad (75)$$

The Newton step $\delta = (\delta E_1, \delta E_2)^\top$ solves $\mathbf{J}\delta = -(f, h)^\top$ via Cramer's rule:

$$\Delta = -g_1 n_1(1 - e_2 \cos E_2) - g_2 n_2(1 - e_1 \cos E_1) \quad (76)$$

$$\delta E_1 = \frac{f n_1(1 - e_2 \cos E_2) + g_2 h}{\Delta} \quad (77)$$

$$\delta E_2 = \frac{-f n_2(1 - e_1 \cos E_1) - g_1 h}{\Delta} \quad (78)$$

where g_1, g_2 are the distance-constraint gradient components (Equation 34, Equation 35) and h is evaluated from Equation 65 (so α enters the Newton step through h). Since $\partial h / \partial E_1 > 0$ and $\partial h / \partial E_2 < 0$ always, the Jacobian is non-singular whenever (g_1, g_2) is not parallel to $(n_2 r_1 / a_1, -n_1 r_2 / a_2)$.

For a hyperbolic spacecraft orbit, the same Cramer's rule formulas apply with $(1 - e_1 \cos E_1)$ replaced by $(e_1 \cosh H_1 - 1)$ (Equation 74), g_1 from Equation 36, and h from Equation 66.

Distance constraint $f = 0$ and time constraints $h = 0$
Voyager 2 — Jupiter, 1970–1990

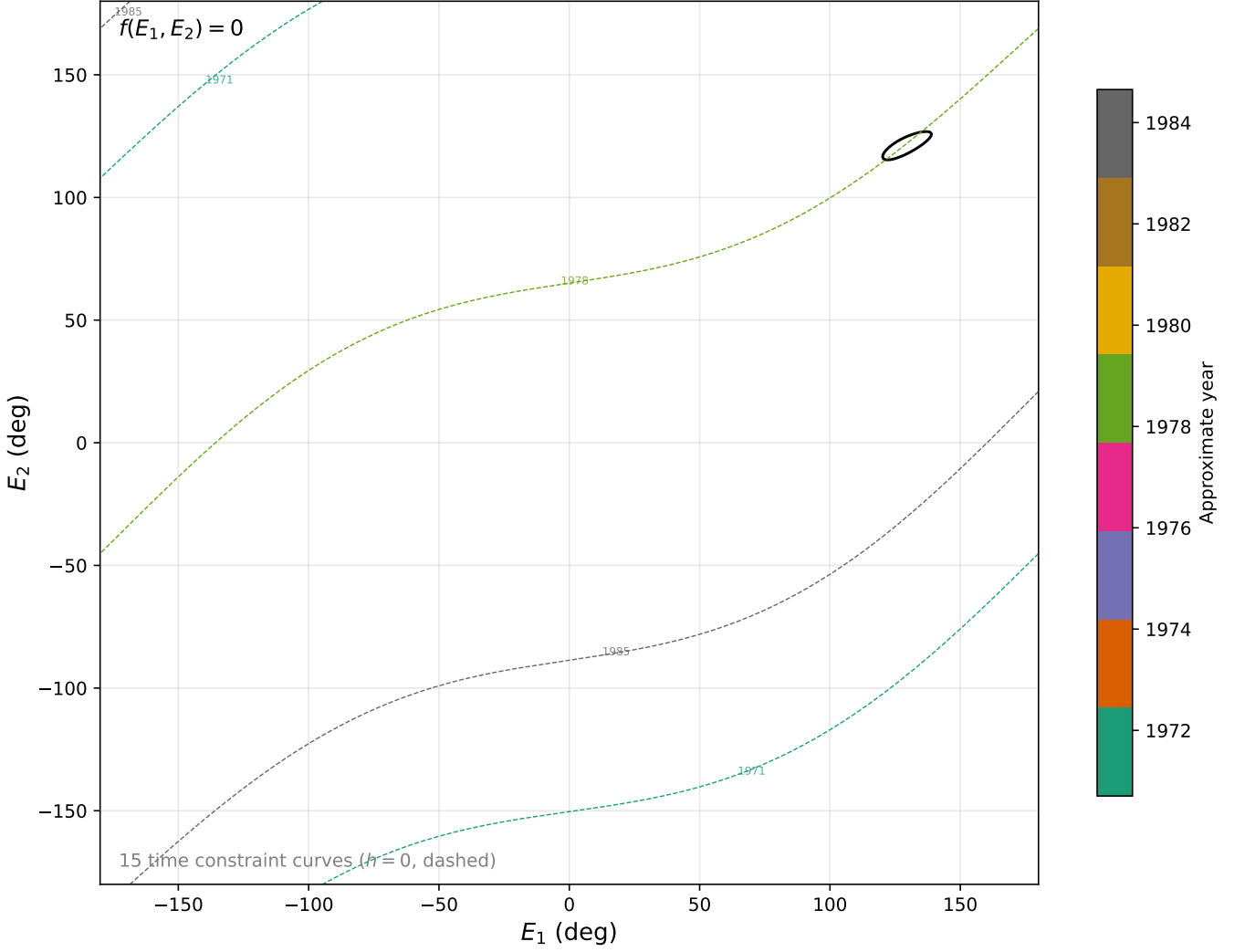


Figure 3: The distance constraint $f(E_1, E_2) = 0$ (solid black) and time constraints $h(E_1, E_2; \alpha) = 0$ (dashed, coloured by year) for the Voyager 2 — Jupiter system (1970–1990). Each dashed curve corresponds to a different encounter opportunity, spaced by the spacecraft’s orbital period ($T_1 \approx 7$ years). Intersections of the dashed curves with the solid curve are solutions to the full encounter problem.

7.6. Initial guess

The unconstrained minimisation of f (from the Newton step in Equation 10) yields the closest-approach point (E_1^*, E_2^*) where $\nabla f = 0$. At this point, evaluate $f_{\min} := f(E_1^*, E_2^*)$ and the Hessian $\mathbf{H}$ (Equation 37, Equation 38, Equation 39).

7.6.1. Quadratic approximation of $f = 0$

The second-order Taylor expansion of a multivariable function around a point $\mathbf{a}$ is:

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a})^\top (\mathbf{x} - \mathbf{a}) + \frac{1}{2} (\mathbf{x} - \mathbf{a})^\top \mathbf{H}(\mathbf{a}) (\mathbf{x} - \mathbf{a}) \quad (79)$$

Let $\mathbf{a} = (E_1^*, E_2^*)$, $\boldsymbol{\delta} = (E_1^* - E_1, E_2^* - E_2)^\top$, and $\mathbf{x} = \mathbf{a} - \boldsymbol{\delta}$. At the minimum, the gradient vanishes ($\nabla f = 0$), so the linear term drops out (the sign of $\boldsymbol{\delta}$ does not matter in the quadratic term):

$$f \approx f_{\min} + \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{H} \boldsymbol{\delta} \quad (80)$$

Setting $f = 0$:

$$\boldsymbol{\delta}^\top \mathbf{H} \boldsymbol{\delta} = -2f_{\min} \quad (81)$$

Written out, the left side is $H_{11}\delta_1^2 + 2H_{12}\delta_1\delta_2 + H_{22}\delta_2^2$ — the general equation of a conic section. Since $\mathbf{H}$ is positive definite at a minimum, this conic is an ellipse. The right side $-2f_{\min} > 0$ when $f_{\min} < 0$ (the orbits come within r_{SOI}), so the ellipse is valid.

7.6.2. Linearised time constraint

The time constraint $h(E_1, E_2; \alpha) = 0$ is linearised around (E_1^*, E_2^*) :

$$h^* - h_1^* \delta_1 - h_2^* \delta_2 = 0 \quad (82)$$

where $h^* = h(E_1^*, E_2^*; \alpha)$, $h_1^* = n_2(1 - e_1 \cos E_1^*)$, $h_2^* = -n_1(1 - e_2 \cos E_2^*)$ (Equation 73). This gives $\delta_2 = c + m\delta_1$ with:

$$c = \frac{h^*}{h_2^*}, \quad m = \frac{h_1^*}{h_2^*} \quad (83)$$

7.6.3. Intersection

Substituting $\delta_2 = c + m\delta_1$ into Equation 81 yields a quadratic in δ_1 :

$$A\delta_1^2 + B\delta_1 + C = 0 \quad (84)$$

with coefficients:

$$A = H_{11} + 2H_{12}m + H_{22}m^2 \quad (85)$$

$$B = 2c(H_{12} + H_{22}m) \quad (86)$$

$$C = H_{22}c^2 + 2f_{\min} \quad (87)$$

The two roots correspond to the two intersections of the time-constraint line with the encounter ellipse. The initial guess is the root with the larger δ_1 (smaller E_1 , earlier encounter):

$$\delta_1 = \frac{-B + \sqrt{B^2 - 4AC}}{2A} \quad (88)$$

giving $(E_1^{(0)}, E_2^{(0)}) = (E_1^* - \delta_1, E_2^* - c - m\delta_1)$. If the discriminant $B^2 - 4AC < 0$, the time-constraint line does not intersect the encounter ellipse for this α , and this encounter opportunity can be skipped.